A single quartic chart for square roots in a quadratic extension

8 November 2026

Abstract

Let F be a number field and let K/F be quadratic. We construct a monic quartic with coefficients polynomial in the two components of an element of K, whose nonzero roots in F recover all its square roots. A cubic change of coordinates ensures that the chosen rational component never vanishes. The proof uses only an element of F that is not a cube. It requires neither a choice among several component charts nor information about rational points on an auxiliary curve. We obtain a polynomial of total degree ten, with one quantified variable over F, testing whether a nonzero element of K is a square. Including zero gives two alternatives: total degree twelve with degree four in the quantified variable, or total degree ten with degree eight in that variable. The chart used over $\mathbb{Q}(\sqrt{2})$ has integer coefficients given explicitly; the latter square test has coefficient height eight.

1 A cubic with no nonzero zero

Let F be a field of characteristic zero, let $d \in F^\times \setminus F^{\times 2}$, and put $K = F(\rho)$ with $\rho^2 = d$. Choose $\kappa \in F^\times \setminus F^{\times 3}$. For an element $u + v\rho$ of K its *rational component* means $u \in F$, irrespective of whether $F = \mathbb{Q}$.

Lemma 1.1. *For $z = u + v\rho \neq 0$, write*

$$z^2 = U + V\rho, \qquad U = u^2 + dv^2, \quad V = 2uv.$$

Define

$$a = U + \frac{\kappa}{2d}V, \qquad b = -\frac{\kappa}{d^2}U - \frac{1}{2}V, \qquad m = a + b\rho. \tag{1}$$

Then

$$mz = (u^3 - \kappa v^3) + \left(dv^3 - \frac{\kappa}{d^2}u^3\right)\rho. \tag{2}$$

In particular the rational component of mz is nonzero. Moreover $m \neq 0$ for every $(U, V) \in F^2 \setminus \{(0, 0)\}$, whether or not $U + V\rho$ is a square.

Proof. The rational component of mz is $au + dbv$. Substitution of $U = u^2 + dv^2$ and $V = 2uv$ gives $u^3 - \kappa v^3$. The same substitution in $av + bu$ gives the other component in (2). If $u^3 = \kappa v^3$, either $v \neq 0$ and $\kappa = (u/v)^3$, or $v = u = 0$. Both are excluded.

The determinant of the linear map $(U, V) \mapsto (a, b)$ is

$$\frac{\kappa^2 - d^3}{2d^3}.$$

It is nonzero, since $\kappa^2 = d^3$ would make $d = (\kappa/d)^2$ a square. As $1, \rho$ are linearly independent over F, the multiplier can vanish only when $a = b = 0$, and therefore only when $U = V = 0$. □

Every number field admits the required choice of κ. Indeed, in its ring of integers choose a nonzero prime ideal $\mathfrak{p}$ and an element $\kappa \in \mathfrak{p} \setminus \mathfrak{p}^2$. Then $v_{\mathfrak{p}}(\kappa) = 1$, whereas the valuation of a cube is divisible by three. Thus Lemma 1.1 applies to every quadratic extension of every number field, with the coefficient κ chosen for that base field.

One may also check the multiplier through its norm:

$$a^2 - db^2 = \frac{d^3 - \kappa^2}{4d^3}(4U^2 - dV^2). \tag{3}$$

Both factors on the right are nonzero for $(U, V) \neq (0, 0)$.

2 The quartic and its inverse

Regard a, b in (1) as linear polynomials in U, V. Define cubic polynomials over F by

$$A = (a^2 + db^2)U + 2dabV, \qquad B = (a^2 + db^2)V + 2abU. \tag{4}$$

They satisfy the exact identity

$$m^2(U + V\rho) = A + B\rho. \tag{5}$$

Put

$$J(U, V, Y) = Y^4 - 4A(U, V)Y^2 + 4dB(U, V)^2. \tag{6}$$

This polynomial is monic of degree four in Y and has total degree at most six.

Theorem 2.1. *Let $(U, V) \in F^2 \setminus \{(0, 0)\}$. Then $U + V\rho$ is a square in K if and only if there exists $Y \in F^\times$ with $J(U, V, Y) = 0$. Every such root recovers a square root by*

$$z = \frac{a - b\rho}{a^2 - db^2}\left(\frac{Y}{2} + \frac{B}{Y}\rho\right). \tag{7}$$

Conversely, every square root z of $U + V\rho$ is recovered by setting Y equal to twice the rational component of mz. Every permitted root Y is simple.

Proof. If $z^2 = U + V\rho$, put $z' = mz = u' + v'\rho$. Lemma 1.1 gives $u' \neq 0$. By (5),

$$(u')^2 + d(v')^2 = A, \qquad 2u'v' = B.$$

Thus $Y = 2u'$ is nonzero, $v' = B/Y$, and $Y^4 - 4AY^2 + 4dB^2 = 0$. The inverse formula follows by dividing z' by m.

Conversely, a nonzero root Y of J satisfies

$$\left(\frac{Y}{2} + \frac{B}{Y}\rho\right)^2 = A + B\rho,$$

because the rational component of the square is $Y^2/4 + dB^2/Y^2 = A$. The multiplier is nonzero by Lemma 1.1, so (5) and (7) give $z^2 = U + V\rho$.

At such a root, write $z' = Y/2 + (B/Y)\rho = u' + v'\rho$. Then

$$\frac{\partial J}{\partial Y} = 4Y^3 - 8AY = 8Y\big((u')^2 - d(v')^2\big) = 8Y\,\mathrm{N}_{K/F}(z') \neq 0.$$

The last inequality follows because $z' = mz$ is a nonzero element of a field. □

Corollary 2.2. *For $(U,V) \neq (0,0)$, the element $U + V\rho$ is a square in K if and only if*

$$1 - 4A(U,V)T^2 + 4dB(U,V)^2T^4 = 0 \tag{8}$$

has a solution $T \in F$. This is a polynomial of total degree at most ten in (U,V,T). Without excluding zero, the equivalent test with one quantified variable is

$$(U^2 - dV^2)(1 - 4A(U,V)T^2 + 4dB(U,V)^2T^4) = 0, \tag{9}$$

of total degree at most twelve.

Proof. Equation (8) excludes $T = 0$, so it is equivalent to (6) under $Y = 1/T$. Since d is not a square, $U^2 - dV^2 = 0$ holds precisely when $U = V = 0$. This proves (9), including the zero square. □

3 A degree-ten test including zero

If the objective is square membership rather than a finite quartic chart with rational reconstruction on one open set, a second formula has lower total degree and smaller coefficients. It does not require a noncube in the base field.

Proposition 3.1. *Let F be any field of characteristic different from two, let $d \in F^\times$ be nonsquare, and let $c \in F^\times$ be any nonsquare, for example $c = d$. An element $U + V\sqrt{d}$ of $F(\sqrt{d})$ is a square, including zero, if and only if there exists $T \in F$ such that*

$$(4 - 4UT^2 + dV^2T^4)(V^2 - c(U - dT^2)^2) = 0. \tag{10}$$

The polynomial has total degree ten and degree eight in T.

Proof. A root of the first factor has $T \neq 0$ and gives

$$\left(\frac{1}{T} + \frac{VT}{2}\sqrt{d}\right)^2 = U + V\sqrt{d}.$$

Conversely, a square root $u + v\sqrt{d}$ with $u \neq 0$ gives such a root by taking $T = 1/u$.

Since c is nonsquare, the second factor vanishes exactly when $V = 0$ and $U = dT^2$. These are precisely the elements with a square root of the form $T\sqrt{d}$, including zero. Thus the two factors cover all square roots, and each factor implies square membership. Their product expresses this alternative using the same quantified variable. □

For $F = \mathbb{Q}$ and $d = 2$, choose $c = -1$ and divide the first factor by two. The test becomes

$$(2 - 2UT^2 + V^2T^4)(V^2 + (U - 2T^2)^2) = 0. \tag{11}$$

It has twelve monomials and coefficient height eight. In contrast, the quartic in Theorem 2.1 has degree four in its finite root and a single rational recovery formula. These are different useful bounds: the lower-height test (11) has degree eight in T and uses two recovery formulas according to the vanishing factor.

4 Integer coefficients over $\mathbb{Q}(\sqrt{2})$

Take $F = \mathbb{Q}$, $d = 2$, and $\kappa = 2$. Multiplying the multiplier in (1) by two gives the simpler integral formulas

$$a = 2U + V, \qquad b = -(U + V),$$
$$A = 6U^3 - 9UV^2 - 4V^3, \qquad B = -4U^3 + 6UV^2 + 3V^3.$$

Thus the explicit monic quartic is

$$Y^4 - 4(6U^3 - 9UV^2 - 4V^3)Y^2 + 8(-4U^3 + 6UV^2 + 3V^3)^2 = 0. \tag{12}$$

The nonzero-root condition and inverse formula in Theorem 2.1 apply with these scaled a, b, A, B. Their elementary identities are

$$(a + b\sqrt{2})(u + v\sqrt{2}) = 2(u^3 - 2v^3) + (4v^3 - u^3)\sqrt{2},$$
$$a^2 - 2b^2 = 2U^2 - V^2,$$
$$(a + b\sqrt{2})^2(U + V\sqrt{2}) = A + B\sqrt{2}.$$

The rational component is nonzero because 2 is not a rational cube. For example, a rational cube has 2-adic valuation divisible by three, whereas $v_2(2) = 1$.

The corresponding one-variable square test, including zero, is

$$(U^2 - 2V^2)\left(1 - 4(6U^3 - 9UV^2 - 4V^3)T^2 + 8(-4U^3 + 6UV^2 + 3V^3)^2T^4\right) = 0.$$

All its coefficients are integers. The monic quartic (12) and the reciprocal test for nonzero squares each have ten monomials and coefficient height 384. The displayed zero-inclusive test has fifteen monomials and coefficient height 1056.

5 Application to direct elliptic charts

Put $D = 11 - 6\sqrt{2}$ and let $E : y^2 = x^3 - Dx$. Its finite points with nonzero square abscissa are represented by

$$x = h^2, \qquad j^2 = h^4 - D, \qquad y = hj.$$

Writing $h = h_0 + h_1\sqrt{2}$ gives

$$U = h_0^4 + 12h_0^2h_1^2 + 4h_1^4 - 11, \qquad V = 4h_0h_1(h_0^2 + 2h_1^2) + 6.$$

The quantity $h^4 - D$ is nonzero: otherwise $h^2 = \pm(3 - \sqrt{2})$; the negative sign is excluded by the real embeddings, and the positive sign by the nonsquare norm 7. Substitution in (12) therefore represents every point in this chart by two rational base coordinates h_0, h_1 and one finite quartic root. The coefficient degrees in the base coordinates are at most twenty-four. Both the nonzero-root condition and its simple-root property are explicit.

The subgroup assertion that these are exactly the nonzero integer multiples of a specified point is a separate arithmetic statement. It follows from the stated group hypothesis and the descent calculation in the companion note *Square abscissas and direct charts for the elliptic subgroup*. The present square-root chart and its number-field extension are unconditional.

Exact verification. The accompanying program `check_quadratic_square_chart.py` verifies the displayed identities, degrees, monomial counts, and coefficient heights by exact polynomial arithmetic over $\mathbb{Q}$, clearing the fixed denominators in d and κ. The nonsquare and noncube hypotheses establish nonvanishing in the proofs above.